

# ICPC Code Template

喵喵喵幼儿园

4 CATEGORIES                      5 TEMPLATES                      95 SOURCE LINES

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# 1 Data Structures

## 1.1 Disjoint Set Union

FILE data-structures/disjoint\_set\_union.cpp

LINES 28

USE Maintain components with path compression and union by size.

COST  $O(\alpha(n))$  amortized per operation

AUTHOR Team Template

```

1 struct DSU {
2     std::vector<int> parent, size;
3
4     explicit DSU(int n) : parent(n), size(n, 1) {
5         std::iota(parent.begin(), parent.end(), 0);
6     }
7
8     int find(int x) {
9         while (x != parent[x]) {
10             parent[x] = parent[parent[x]];
11             x = parent[x];
12         }
13         return x;
14     }
15
16     bool unite(int a, int b) {
17         a = find(a);
18         b = find(b);
19         if (a == b) return false;
20         if (size[a] < size[b]) std::swap(a, b);
21         parent[b] = a;
22         size[a] += size[b];
23         return true;
24     }
25
26     bool same(int a, int b) { return find(a) == find(b); }
27     int component_size(int x) { return size[find(x)]; }
28 };

```

## 2 Graph

### 2.1 Dijkstra Shortest Paths

FILE graph/dijkstra.cpp

LINES 30

USE Single-source shortest paths in a graph with non-negative edge weights.

COST  $O((n + m) \log n)$ 

AUTHOR Team Template

```

1 using i64 = long long;
2 constexpr i64 INF = std::numeric_limits<i64>::max() / 4;
3
4 struct Edge {
5     int to;
6     i64 weight;
7 };
8
9 std::vector<i64> dijkstra(const std::vector<std::vector<Edge>>& graph, int source) {
10     const int n = static_cast<int>(graph.size());
11     std::vector<i64> distance(n, INF);
12     using State = std::pair<i64, int>;
13     std::priority_queue<State, std::vector<State>, std::greater<State>> queue;
14
15     distance[source] = 0;
16     queue.emplace(0, source);
17     while (!queue.empty()) {
18         auto [current_distance, vertex] = queue.top();
19         queue.pop();
20         if (current_distance != distance[vertex]) continue;
21
22         for (const Edge& edge : graph[vertex]) {
23             const i64 candidate = current_distance + edge.weight;
24             if (candidate >= distance[edge.to]) continue;
25             distance[edge.to] = candidate;
26             queue.emplace(candidate, edge.to);
27         }
28     }
29     return distance;
30 }

```

## 3 Math

### 3.1 Binary GCD

FILE math/binary\_gcd.cpp

LINES 6

USE 超快 gcd 算法

COST  $O(\log n)$ , 可以当作  $O(1)$ 

AUTHOR ppip

```

1 #define ctz __builtin_ctz
2 inline int gcd(int a,int b) {
3     int az{ctz(a)},bz{ctz(b)},z{min(az,bz)},t;b>>=bz;
4     while (a) a>>=az,t=a-b,az=ctz(t),b=min(a,b),a=abs(t);
5     return b<<z;
6 }
```

### 3.2 Modular Power and Inverse

FILE math/modular\_power.cpp

LINES 16

USE Binary exponentiation and multiplicative inverse modulo a prime.

COST  $O(\log \text{exponent})$ 

AUTHOR Team Template

```

1 using i64 = long long;
2
3 i64 modular_power(i64 base, i64 exponent, i64 modulus) {
4     base %= modulus;
5     i64 result = 1 % modulus;
6     while (exponent > 0) {
7         if (exponent & 1) result = static_cast<__int128>(result) * base % modulus;
8         base = static_cast<__int128>(base) * base % modulus;
9         exponent >>= 1;
10    }
11    return result;
12 }
13
14 i64 modular_inverse_prime(i64 value, i64 prime_modulus) {
15     return modular_power(value, prime_modulus - 2, prime_modulus);
16 }
```

## 4 Misc

### 4.1 快速模乘

FILE misc/fastmod.cpp

LINES 15

USE 要求  $1 \leq P < 2^{31}$  且  $a < P$ ; 复用  $\text{trans}(z)$  时第一个 mul 更快COST  $O(1)$ 

AUTHOR ppip

```

1 using u32=unsigned;
2 using u64=unsigned long long;
3 using u128=unsigned __int128;
4 const u32 P=998244353;
5 inline u64 trans(u64 x) {
6     constexpr u64 A=-(u64)P/P+1;
7     constexpr u64 q=((u128(-(u64)P%P)<<64)+P-1)/P;
8     return x*A+u64((u128)x*q>>64)+1;
9 }
10 inline void mul(u64& a,u64 z) {
11     a=a*z*(u128)P>>64;
12 }
13 inline u32 mul(u32 a,u64 z) {
14     return a*trans(z)*(u128)P>>64;
15 }
```